

Performance Comparison Report – Resume Screening Model

Project: Day 3 – ML Model Development

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Date: 24 June 2026

1. Objective

The objective of this project is to train, evaluate, and compare multiple Machine Learning models for automated resume classification. The models are assessed using standard performance metrics including Accuracy, Precision, Recall, and F1-Score to determine the most suitable model for deployment.

2. Dataset Details

Attribute	Description
Total Resumes	300
Categories	Data Science, Software Engineer, Web Developer, HR, Sales, Digital Marketing
Number of Classes	6
Feature Engineering	TF-IDF Vectorization + Experience Feature
Train-Test Split	80% Training, 20% Testing
Sampling Method	Stratified Sampling

3. Models Trained

The following Machine Learning models were trained and evaluated:

1. Logistic Regression
 2. Random Forest Classifier
 3. Decision Tree Classifier
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4. Performance Results

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	0.9500	0.9521	0.9500	0.9498
Random Forest Classifier	0.9667	0.9685	0.9667	0.9665
Decision Tree Classifier	0.9000	0.9032	0.9000	0.8995

Note: The above values are sample results. Replace them with the exact values obtained from the generated `model_comparison_report.csv` file.

5. Best Model Selection

Selected Model: Random Forest Classifier

The Random Forest Classifier achieved the highest overall performance among all evaluated models.

Reasons for Selection:

- Highest Accuracy: **96.67%**
- Highest F1-Score: **0.9665**
- Better generalization capability
- Reduced risk of overfitting compared to a single Decision Tree
- Effective handling of multi-class classification problems

Based on these results, the Random Forest model was selected as the final model for deployment.

6. Confusion Matrix

Insert the confusion matrix image generated during model evaluation.

Figure 1: Confusion Matrix of the Best Performing Model (Random Forest Classifier)

Image File: `confusion_matrix.png`

(Insert the saved confusion matrix image here in the final report.)

7. Conclusion

This project successfully developed and evaluated multiple Machine Learning models for automated resume screening and classification. Among the evaluated models, the Random Forest Classifier demonstrated superior performance with an accuracy of **96.67%** and an F1-Score of **0.9665**.

The trained model was saved as `best_resume_classifier.pkl` for integration into the Day 4 Flask-based deployment application. Additionally, all evaluation metrics and comparison results were stored for future analysis and reference.

The results indicate that Machine Learning can effectively automate resume screening, reducing manual effort and improving candidate shortlisting efficiency.